

	EMERGENCY MEDICAL SERVICES AGENCY	POLICY NUMBER	708
	A Division of the Merced County Department of Public Health	Effective Date	11/2025
Authority: Health and Safety Code, Division 2.5, and California Code of Regulations, Title 22, Division 9, Chapter 4, 1797.220		Initial Date:	07/2022
		Review Date:	10/2028

TACHYARRHYTHMIAS	
ADULT	PEDIATRIC
BLS	
Assess Vitals / Obtain SpO2 / Oxygen - Titrate to SpO2 of 94% or higher / Secure Airway - Assist Ventilations if needed.	
ALS	
Follow BLS procedures if applicable. Obtain 4-Lead ECG and 12-Lead ECG (12-Lead for Adult Only). Vascular Access - IV/IO rate as appropriate. Capnography.	
Symptomatic Tachycardia	
Symptomatic Tachycardia is defined by ALL the following: Heart rate above: 150 BPM - Adult 220 BPM - Infant (less than 2 years of age) 180 BPM - Toddler (ages 2 years and greater) Systolic Blood Pressure: - Adult - Below 90 mmHg. - Pediatric - Hypotensive verified by length-based assessment tape or application. Associated signs and symptoms including: - ALOC, - signs of shock, - chest pain, - acute pulmonary edema, - syncope, extreme weakness.	
Sinus Tachycardia	
Consider IV/IO fluid bolus and look for underlying causes (dehydration, sepsis, shock) – Refer to EMS Policy #738 – Non-Traumatic Shock.	

Signatures on File

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Supraventricular Tachycardia (SVT)

With no associated signs or symptoms of extremis:

VALSALVA MANEUVER - If heart rate greater than 150 BPM.

ADENOSINE - 6 mg rapid IV push over 1 second followed by 10-20 ml saline flush

- Repeat with **12 mg** rapid IV push over 1 second followed by 10-20 ml saline flush, if patient does not convert in 2 minutes
- Repeat with **12 mg** rapid IV push over 1 second followed by 10-20 ml saline flush, if patient does not convert in 1 minute

With no associated signs or symptoms of extremis:

Consider IV/IO fluid bolus and look for underlying causes (dehydration, sepsis, shock) – **Refer to EMS Policy #738 – Non-Traumatic Shock.**

If the patient has narrow complex tachycardia with mild symptoms and no response to IV/IO fluids, consider Adenosine.

Base Hospital Physician Orders Only

With no associated signs or symptoms of extremis:

ADENOSINE - 0.1 mg/kg, max dose 6 mg, rapid IV push over 1 second followed by 10-20 ml saline flush

- Repeat with **0.2 mg/kg, max dose 12 mg, rapid IV push over 1 second followed by 10-20 ml saline flush, if patient does not convert**

With associated signs or symptoms of extremis:

SYNCHRONIZED CARDIOVERSION – Per manufacturers recommended setting.

- If the patient is unstable perform immediate synchronized cardioversion

Ventricular Tachycardia (VT)

With no associated signs or symptoms of extremis:

Consider IV/IO fluid bolus and look for underlying causes (dehydration, sepsis, shock) – **Refer to EMS Policy #738 – Non-Traumatic Shock.**

With associated signs or symptoms of extremis:

SYNCHRONIZED CARDIOVERSION – Per manufacturers recommended setting.

- If the patient is unstable perform immediate synchronized cardioversion

Successful Cardioversion – Amiodarone OR Lidocaine AND Magnesium Sulfate:

Amiodarone - 150 mg infusion over 10 minutes

Lidocaine - 0.5 mg/kg

- May repeat every 10 minutes to maximum dose of 3 mg/kg

Magnesium Sulfate - 2 gms in 10 ml saline slow IV push over 2 minutes

No Medication Orders

Torsades De Pointe Polymorphic (Wide Complex)

With no associated signs or symptoms of extremis:

Magnesium Sulfate - 2 gms in 10 ml saline slow IV push over 2 minutes.

With associated signs or symptoms of extremis:

Manual Defibrillation – Per manufacturers recommended setting.

NO STANDING ORDERS

Atrial Fibrillation

With associated signs or symptoms of extremis:

SYNCHRONIZED CARDIOVERSION – Per manufacturers recommended setting.

- If heart rate greater than 150 BPM, and the patient is unstable, perform immediate synchronized cardioversion

NO STANDING ORDERS

Synchronized Cardioversion Sedation / Pain Management

Sedation:

Midazolam - 2-5 mg IV/IO

- Single dose

Sedation:

Midazolam - **0.1 mg/kg IV/IO, max dose 2 mg**

- Single dose

Pain Management – Morphine OR Fentanyl:

Morphine - 2-5 mg IV/IO

- May repeat once after five (5) minutes if SBP > 90 mmHg

Fentanyl - 25 - 50 mcg IV/IO

- May repeat once after five (5) minutes if SBP > 90 mmHg

Fentanyl or Morphine is preferred for patients with chest pain or suspected STEMI's.

Pain Management – Morphine OR Fentanyl:

Morphine - 0.1 mg/kg IV/IO, max dose 2.5 mg

- May repeat once after 5 minutes if SBP is greater than length-based assessment tape or application target

Fentanyl - 1 mcg/kg IV/IO, max dose 25 mcg

- May repeat once after 5 minutes if SBP is greater than length-based assessment tape or application target

Special Considerations

If at any time the monitor will not synchronize with the rhythm, and the patient is unstable, perform immediate defibrillation.